Chapter 6. The 2^k Factorial Design

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1 The 2^2 Factorial Design

- The 2^k design is a special case of the general factorial: k factors, each at **two levels** ("low" $-$ and "high" $+$; quantitative or qualitative). Very widely used in industry (screening).
- Simplest case: 2^2. Effects model

$$y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + \varepsilon_{ijk}, \qquad i = 1, 2;\ j = 1, 2;\ k = 1, \ldots, n.$$

1.1 Notation for treatment combinations

A treatment combination is denoted by the lowercase letters of the factors at their high level: (1) = both low, $a = A$ high, $b = B$ high, ab = both high. The same symbol also denotes the **total** of the n observations at that combination.

$$\begin{array}{ccc} b\ (-,+) & — & ab\ (+,+) \\ | & & | \\ (1)\ (-,-) & — & a\ (+,-) \end{array}$$

Example 6.1: Chemical process data ($n = 3$)

A = reactant concentration, B = amount of catalyst, response = yield.

A	B	I	II	III	Total
$-$	$-$	28	25	27	$(1) = 80$
$+$	$-$	36	32	32	$a = 100$
$-$	$+$	18	19	23	$b = 60$
$+$	$+$	31	30	29	$ab = 90$

Analysis procedure: (1) estimate factor effects (main effects A, B, interaction AB); (2) statistical testing (ANOVA: SS for A, B, AB, error); (3) residual analysis (normality, constant variance).

1.2 Estimating effects

Derivation

The effect of A at the low level of B is $\dfrac{a-(1)}{n}$; at the high level of B it is $\dfrac{ab-b}{n}$. The main effect of A is their average:

$$A = \frac{1}{2}\Big(\frac{a-(1)}{n} + \frac{ab-b}{n}\Big) = \frac{ab+a-b-(1)}{2n}.$$

Equivalently $A = \bar{y}_{A^+} - \bar{y}_{A^-} = \dfrac{ab+a}{2n} - \dfrac{b+(1)}{2n}$.

$$A = \frac{1}{2n}[ab+a-b-(1)], \qquad B = \frac{1}{2n}[ab+b-a-(1)], \qquad AB = \frac{1}{2n}[ab+(1)-a-b].$$

AB is half the difference between the A effect at high B and at low B: $\frac{1}{2}\big[\frac{ab-b}{n} - \frac{a-(1)}{n}\big]$.

Table of algebraic signs

Treatment combination	I	A	B	AB
(1)	+	−	−	+
a	+	+	−	−
b	+	−	+	−
ab	+	+	+	+

The AB column is the elementwise product of the A and B columns. Every pair of columns is orthogonal.

1.3 Sums of squares via contrasts

Theorem 6.2: SS of a contrast

For a contrast $C = \sum_\ell c_\ell S_\ell$ in the treatment totals S_ℓ (each a sum of n observations), $\sum_\ell c_\ell = 0$,

$$SS_C = \frac{(\text{contrast})^2}{n\sum_\ell c_\ell^2}.$$

In the 2^2 design: $\text{Contrast}_A = ab+a-b-(1)$, $\text{Contrast}_B = ab+b-a-(1)$, $\text{Contrast}_{AB} = ab+(1)-a-b$, $\sum c_\ell^2 = 4$, so

$$SS_A = \frac{[ab+a-b-(1)]^2}{4n}, \quad SS_B = \frac{[ab+b-a-(1)]^2}{4n}, \quad SS_{AB} = \frac{[ab+(1)-a-b]^2}{4n}.$$

Direct proof for SS_A. In the two-factor notation $SS_A = \sum_{i=1}^{2}\sum_{j=1}^{2}\sum_{k=1}^{n}(\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2 = 2n\sum_{i=1}^{2}(\bar{y}_{i\cdot\cdot}-\bar{y}_{\cdots})^2 = 2n\big[(\bar{y}_{A^+}-\bar{y}_{\cdots})^2+(\bar{y}_{A^-}-\bar{y}_{\cdots})^2\big]$, with $\bar{y}_{A^+} = \frac{ab+a}{2n}$, $\bar{y}_{A^-} = \frac{b+(1)}{2n}$, $\bar{y}_{\cdots} = \frac{ab+a+b+(1)}{4n}$:

$$\bar{y}_{A^+} - \bar{y}_{\cdots} = \frac{2(ab+a)-(ab+a+b+(1))}{4n} = \frac{ab+a-b-(1)}{4n}, \qquad \bar{y}_{A^-} - \bar{y}_{\cdots} = -\frac{ab+a-b-(1)}{4n},$$

$$SS_A = 2n\cdot 2\cdot\frac{[ab+a-b-(1)]^2}{16n^2} = \frac{[ab+a-b-(1)]^2}{4n}.$$

■

General: $t_0^2 = F_0$. Let $S_\ell = \sum_{k=1}^n y_{\ell k} = n\bar{y}_\ell$ (ℓ indexes the cells), and $C = \sum_\ell c_\ell S_\ell = n\sum_\ell c_\ell \bar{y}_\ell$. Then

$$\mathrm{Var}(C) = \mathrm{Var}\Big(n\sum_\ell c_\ell \bar{y}_\ell\Big) = n^2 \sum_\ell c_\ell^2 \mathrm{Var}(\bar{y}_\ell) = n^2 \sum_\ell c_\ell^2 \frac{\sigma^2}{n} = n\sigma^2 \sum_\ell c_\ell^2.$$

Let $t_0 = \dfrac{C - \mathrm{E}(C)}{\sqrt{\widehat{\mathrm{Var}}(C)}} = \dfrac{\sum c_\ell S_\ell - \mathrm{E}(C)}{\sqrt{n\, MS_E \sum c_\ell^2}}$. Assuming $\mathrm{E}(C) = 0$ (true under H_0, see below),

$$t_0^2 = \frac{(\sum_\ell c_\ell S_\ell)^2/(n\sum c_\ell^2)}{MS_E} = \frac{SS_C/1}{MS_E} \sim F_{1,\,2^2(n-1)}.$$

So each effect's F-test in the ANOVA is the square of a t-test of the contrast. ■

Derivation: $\mathrm{E}(\mathrm{Contrast}_A) = 0$ under H_0

With effects A_1, A_2 (low/high) etc.,

$$ab = \textstyle\sum_k y_{22k} = n(\mu + A_2 + B_2 + (AB)_{22}) + \varepsilon_{22\cdot}, \qquad a = n(\mu + A_2 + B_1 + (AB)_{21}) + \varepsilon_{21\cdot},$$
$$b = n(\mu + A_1 + B_2 + (AB)_{12}) + \varepsilon_{12\cdot}, \qquad (1) = n(\mu + A_1 + B_1 + (AB)_{11}) + \varepsilon_{11\cdot}.$$

Hence

$$\mathrm{E}(\mathrm{Contrast}_A) = n(2A_2 - 2A_1 + \underbrace{(AB)_{22} + (AB)_{21} - (AB)_{12} - (AB)_{11}}_{=0 \text{ by } \sum_i (AB)_{ij} = 0}) = n(2A_2 - 2A_1) \overset{H_0}{=} 0.$$

Since $A_1 + A_2 = 0$, $\mathrm{E}(\mathrm{Contrast}_A) = 4nA_2$ and $\mathrm{E}(A \text{ effect}) = \frac{4nA_2}{2n} = 2A_2 = A_2 - A_1$.

1.4 ANOVA

DF: $2^2 n - 1 = 1 + 1 + 1 + 2^2(n-1)$.

Source	SS	DF	MS	F_0
A	SS_A	1	MS_A	MS_A/MS_E
B	SS_B	1	MS_B	MS_B/MS_E
AB	SS_{AB}	1	MS_{AB}	MS_{AB}/MS_E
Error	SS_E	$2^2(n-1)$	MS_E	
Total	SS_T	$2^2 n - 1$		

Example 6.3: Chemical process, line by line

$$A = \frac{90 + 100 - 60 - 80}{2(3)} = \frac{50}{6} = 8.33,$$
$$B = \frac{90 + 60 - 100 - 80}{6} = \frac{-30}{6} = -5.00,$$
$$AB = \frac{90 + 80 - 100 - 60}{6} = \frac{10}{6} = 1.67,$$
$$SS_A = \frac{50^2}{4(3)} = 208.33, \quad SS_B = \frac{(-30)^2}{12} = 75.00, \quad SS_{AB} = \frac{10^2}{12} = 8.33,$$
$$SS_T = \sum y^2 - \frac{y_{\cdots}^2}{12} = 9398 - \frac{330^2}{12} = 9398 - 9075 = 323.00,$$

$SS_E = 323.00 - 208.33 - 75.00 - 8.33 = 31.34.$

Source	SS	DF	MS	F_0
A	208.33	1	208.33	53.15
B	75.00	1	75.00	19.13
AB	8.33	1	8.33	2.13
Error	31.34	8	3.92	
Total	323.00	11		

$F_{0.05,1,8} = 5.32$: A and B significant, AB not. Increasing concentration raises yield; increasing catalyst lowers it.

2 The 2^3 Design

Factors A, B, C at two levels: 8 runs $(1), a, b, ab, c, ac, bc, abc$.

Trt	I	A	B	AB	C	AC	BC	ABC
(1)	+	−	−	+	−	+	+	−
a	+	+	−	−	−	−	+	+
b	+	−	+	−	−	+	−	+
ab	+	+	+	+	−	−	−	−
c	+	−	−	+	+	−	−	+
ac	+	+	−	−	+	+	−	−
bc	+	−	+	−	+	−	+	−
abc	+	+	+	+	+	+	+	+

Properties: (1) except for I, every column has equally many + and − (a contrast); (2) the product of any two columns is a column of the table ($A \cdot B = AB$, $AB \cdot BC = AB^2C = AC$, with squares $= I$); (3) I times any column leaves it unchanged; (4) all columns are mutually orthogonal.

$$A = \frac{1}{4n}[(abc + ac + ab + a) - (bc + c + b + (1))], \qquad SS_A = \frac{(\text{Contrast}_A)^2}{8n}.$$

Theorem 6.4: General 2^k

$$\text{Effect} = \frac{\text{Contrast}}{2^{k-1}n}, \qquad SS = \frac{(\text{Contrast})^2}{2^k n}, \qquad \text{Var(Effect)} = \frac{2^k n\sigma^2}{(2^{k-1}n)^2} = \frac{\sigma^2}{2^{k-2}n},$$

$\text{df(Total)} = n2^k - 1$, $\text{df(Error)} = 2^k(n-1)$, each of the $2^k - 1$ effects has 1 df.

Proof. Each contrast uses all 2^k totals with $c_\ell = \pm 1$, so $\sum c_\ell^2 = 2^k$ and $SS = \text{Contrast}^2/(n2^k)$. The effect is (sum of 2^{k-1} totals at +)$/(2^{k-1}n)$ minus (sum at −)$/(2^{k-1}n)$, i.e. $\text{Contrast}/(2^{k-1}n)$. Variance: $\text{Var(Contrast)} = n\sigma^2 \sum c_\ell^2 = 2^k n\sigma^2$. ■

Source	SS	DF
A, B, C	SS_A, SS_B, SS_C	1 each
AB, AC, BC	$SS_{AB}, SS_{AC}, SS_{BC}$	1 each
ABC	SS_{ABC}	1
Error	SS_E	$2^3(n-1)$
Total	SS_T	$n2^3 - 1$

Example 6.5: A 2^3 with $n = 2$ (soft-drink fill height)

Totals: $(1) = -4$, $a = 1$, $b = -1$, $ab = 5$, $c = -1$, $ac = 3$, $bc = 2$, $abc = 11$ (sum $= 16$).

$$\begin{aligned}
\text{Contrast}_A &= a + ab + ac + abc - (1) - b - c - bc = 1 + 5 + 3 + 11 + 4 + 1 + 1 - 2 = 24,\\
\text{Contrast}_B &= b + ab + bc + abc - (1) - a - c - ac = -1 + 5 + 2 + 11 + 4 - 1 + 1 - 3 = 18,\\
\text{Contrast}_C &= c + ac + bc + abc - (1) - a - b - ab = -1 + 3 + 2 + 11 + 4 - 1 + 1 - 5 = 14,\\
\text{Contrast}_{AB} &= (1) + ab + c + abc - a - b - ac - bc = -4 + 5 - 1 + 11 - 1 + 1 - 3 - 2 = 6,\\
\text{Contrast}_{AC} &= (1) + b + ac + abc - a - ab - c - bc = -4 - 1 + 3 + 11 - 1 - 5 + 1 - 2 = 2,\\
\text{Contrast}_{BC} &= (1) + a + bc + abc - b - ab - c - ac = -4 + 1 + 2 + 11 + 1 - 5 + 1 - 3 = 4,\\
\text{Contrast}_{ABC} &= a + b + c + abc - (1) - ab - ac - bc = 1 - 1 - 1 + 11 + 4 - 5 - 3 - 2 = 4.
\end{aligned}$$

Effects $=$ Contrast$/(4\cdot 2)$: $A = 3.00$, $B = 2.25$, $C = 1.75$, $AB = 0.75$, $AC = 0.25$, $BC = 0.50$, $ABC = 0.50$. SS $=$ Contrast$^2/16$: 36.00, 20.25, 12.25, 2.25, 0.25, 1.00, 1.00. With $SS_T = 78.00$, $SS_E = 78 - 73 = 5.00$ on 8 df, $MS_E = 0.625$; $F_A = 57.6$, $F_B = 32.4$, $F_C = 19.6$, $F_{AB} = 3.6$ ($F_{0.05,1,8} = 5.32$): only the main effects are significant.

3 Regression Approach to the 2^k Design

Derivation

2^2: $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_{12} x_{1i} x_{2i} + \varepsilon_i$ with coded $x = \pm 1$. Using the totals as responses,

$$\mathbf{X} = \begin{pmatrix} 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} (1) \\ a \\ b \\ ab \end{pmatrix}$$

$$\Rightarrow \ \mathbf{X}^T\mathbf{y} = \begin{pmatrix} (1) + a + b + ab \\ -(1) + a - b + ab \\ -(1) - a + b + ab \\ (1) - a - b + ab \end{pmatrix} = \begin{pmatrix} \text{Contrast}_{\text{total}} \\ \text{Contrast}_A \\ \text{Contrast}_B \\ \text{Contrast}_{AB} \end{pmatrix}.$$

Since $\mathbf{X}^T\mathbf{X} = 4\mathbf{I}$, $\hat{\boldsymbol{\beta}} = \frac{1}{4}\mathbf{X}^T\mathbf{y}$, e.g.

$$\hat{\beta}_1 = \tfrac{1}{4}\text{Contrast}_A = \tfrac{1}{4}(2n \cdot \text{main effect } A) = \tfrac{n}{2} \cdot \text{main effect } A.$$

2^3: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \beta_3 x_3 + \beta_{13} x_1 x_3 + \beta_{23} x_2 x_3 + \beta_{123} x_1 x_2 x_3 + \varepsilon$, $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = \frac{1}{2^3}(\text{Contrast}_{\text{total}}, \text{Contrast}_A, \dots)^T$ and since $A = \frac{1}{2^{k-1}n}\text{Contrast}_A$, $\hat{\beta}_1 = \frac{n}{2}\,\text{main effect}_A$.

Remark 6.6: Using cell averages instead

If the responses are the cell *averages* $\bar{y}_\ell = S_\ell/n$, then $\hat{\beta}_1 = \frac{1}{4}\sum_\ell c_\ell \bar{y}_\ell = \frac{1}{4n}\text{Contrast}_A = \frac{A}{2}$: a regression coefficient is **half the effect**, because x moves from -1 to $+1$ (two units). Chemical data: $\mathbf{y} = (80/3, 100/3, 60/3, 90/3)^T$ gives $\hat{\beta}_0 = 27.5$, $\hat{\beta}_1 = 4.1667 = 8.33/2$, $\hat{\beta}_2 = -2.5 = -5/2$, $\hat{\beta}_{12} = 0.8333 = 1.67/2$ (matching the R output).

4 Residual Analysis

Fitted values from the regression model (only significant terms): chemical process $\hat{y} = 27.5 + \frac{8.33}{2}x_1 - \frac{5.00}{2}x_2$. Check normality (Q–Q plot of residuals) and constant variance (residuals vs fitted).

5 Summary of Designs So Far

Design	Model	Analysis
CRD	$y_{ij} = \mu + \tau_i + \varepsilon_{ij}$	one-way ANOVA / regression
RCBD, BIBD	$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$	two-way ANOVA / regression
Latin square	$y_{ijk} = \mu + \alpha_i + \tau_j + \beta_k + \varepsilon_{ijk}$	three-way ANOVA
Graeco-Latin	$y_{ijkl} = \mu + \theta_i + \tau_j + \omega_k + \Psi_l + \varepsilon_{ijkl}$	four-way ANOVA
Factorial	$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$	two-way ANOVA with interaction
2^k	contrasts of $(1), a, b, ab, \ldots$	ANOVA via contrasts / regression